

The T0 Energy Field Model: Mathematical Formulation

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Abstract

The T0 model describes physical phenomena through a universal energy field $E_{\text{field}}(x, t)$ with the parameter $\xi = \frac{4}{3} \times 10^{-4}$. The field equation is $\square E_{\text{field}} = 0$, the Lagrangian density $\mathcal{L} = \xi(\partial E)^2$. The model uses standard natural units with $\hbar = c = 1$.

Fundamental quantities:

- Characteristic energy: $E_0 = \sqrt{m_e \cdot m_\mu} = 7.348 \text{ MeV}$
- Fine structure constant: $\alpha = \xi(E_0/1 \text{ MeV})^2 \approx 1/137$
- Gravitational constant: $G = \xi^2/(4m_e) \times \text{factors}$

Predictions: Lepton masses with 2% accuracy, anomalous magnetic moments $a_\ell = \frac{\xi}{2\pi}(E_\ell/E_e)^2$, fine structure constant with 0.03% agreement.

Detailed derivations: See Document 011 (fine structure), 012 (gravitation), 018 (g-2 geometric), 019 (Lagrangian).

1 Units Convention

T0 natural units (Heaviside-Lorentz)

T0 uses Heaviside-Lorentz natural units:

$$\hbar = c = 4\pi\epsilon_0 = 1 \quad (1)$$

In this system:

- Fine structure constant: $\alpha = e^2 = 1$ (by convention in T0)
- Energy = Mass: $E = m$
- Length = Time = Energy $^{-1}$: $[L] = [T] = [E^{-1}]$

Note: In SI units, $\alpha \approx 1/137$. The choice $\alpha = 1$ in T0 is a unit convention that simplifies formulas.

Dimensions in natural units

$$[E] = E \quad (2)$$

$$[m] = E \quad (3)$$

$$[t] = E^{-1} \quad (4)$$

$$[L] = E^{-1} \quad (5)$$

$$[G] = E^{-2} \quad (6)$$

$$[\partial_\mu] = E \quad (7)$$

.2 Time-Energy Duality

Fundamental relation

$$T_{\text{field}}(x, t) \cdot E_{\text{field}}(x, t) = 1 \quad (8)$$

with $[T_{\text{field}}] = E^{-1}$ and $[E_{\text{field}}] = E$.

Intrinsic time field

$$T_{\text{field}}(x, t) = \frac{1}{E_{\text{field}}(x, t)} \quad (9)$$

.3 Universal Field Equation

Wave equation

$$\square E_{\text{field}} = 0 \quad (10)$$

with d'Alembert operator:

$$\square = \nabla^2 - \frac{\partial^2}{\partial t^2} \quad (11)$$

With sources

$$\nabla^2 E_{\text{field}} = 4\pi G\rho \cdot E_{\text{field}} \quad (12)$$

Dimension check: $[E^3] = [E^{-2}][E^4][E] = [E^3] \checkmark$

.4 Lagrangian Density

Universal Lagrangian density

$$\mathcal{L} = \xi \cdot (\partial_\mu E_{\text{field}})(\partial^\mu E_{\text{field}}) \quad (13)$$

with $\xi = \frac{4}{3} \times 10^{-4}$.

Euler-Lagrange equation

$$\frac{\partial \mathcal{L}}{\partial E} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu E)} = 0 \quad (14)$$

yields:

$$\square E_{\text{field}} = 0 \quad (15)$$

.5 Characteristic Energy

Definition

The characteristic energy E_0 is the geometric mean of electron and muon mass (derivation in Document 011):

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (16)$$

Numerical values

From experimental masses:

$$E_0 = \sqrt{0.511 \times 105.66} \quad (17)$$

$$= \sqrt{53.99} \quad (18)$$

$$= 7.348 \text{ MeV} \quad (19)$$

Theoretical T0 value:

$$E_0^{\text{T0}} = 7.398 \text{ MeV} \quad (20)$$

Deviation: 0.7% (within geometric corrections)

Usage

E_0 serves as energy scale for:

- Fine structure constant: $\alpha = \xi(E_0/1 \text{ MeV})^2$
- Normalization of electromagnetic effects
- Scaling of anomalous magnetic moments

.6 The Parameter ξ

Definition

$$\xi = \frac{4}{3} \times 10^{-4} = 1.3333 \times 10^{-4} \quad (21)$$

Dimensionless: $[\xi] = 1$.

Geometric components

$$\xi = G_3 \times S_{\text{ratio}} \quad (22)$$

where:

- $G_3 = \frac{4}{3}$: Geometric factor (sphere-cube ratio)
- $S_{\text{ratio}} = 10^{-4}$: Scale ratio

.7 Fine Structure Constant

In T0 units

In T0 natural units: $\alpha = 1$ (by convention)

Reconstruction of SI value

Important: The SI value can be reconstructed from T0 parameters!

$$\alpha_{\text{SI}} = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2 \quad (23)$$

Numerical calculation

With $\xi = \frac{4}{3} \times 10^{-4}$ and $E_0 = 7.398 \text{ MeV}$:

$$\alpha_{\text{SI}} = 1.3333 \times 10^{-4} \times (7.398)^2 \quad (24)$$

$$= 1.3333 \times 10^{-4} \times 54.73 \quad (25)$$

$$= 7.297 \times 10^{-3} \quad (26)$$

$$= \frac{1}{137.04} \quad (27)$$

$$\text{Experimental: } \alpha_{\text{exp}} = \frac{1}{137.036}$$

Agreement: 0.03%

Dimension check

$$[\alpha_{\text{SI}}] = [\xi] \times \left[\frac{E}{E} \right]^2 = 1 \times 1 = 1 \quad (28)$$

.8 Gravitational Constant

TO formula

The gravitational constant is derived from ξ and m_e (derivation in Document 012):

$$G = \frac{\xi^2}{4m_e} \times C_{\text{dim}} \times C_{\text{conv}} \quad (29)$$

where:

- C_{dim} : Dimension correction
- C_{conv} : SI conversion factor

Fundamental relation

In natural units:

$$\xi = 2\sqrt{G \cdot m_e} \quad (30)$$

Solved for G :

$$G_{\text{nat}} = \frac{\xi^2}{4m_e} \quad (31)$$

Dimension: $[G] = [E^{-2}]$ in natural units.

.9 Characteristic Lengths

T0 characteristic length

$$r_0 = 2GE \quad (32)$$

$$\text{Dimension: } [r_0] = [E^{-2}][E] = [E^{-1}] = [L] \quad \checkmark$$

Derivation

For spherically symmetric point source $\rho(r) = E_0 \delta^3(\vec{r})$:

Solution of $\nabla^2 E = 4\pi G \rho E$:

$$E(r) = E_0 \left(1 - \frac{r_0}{r} \right) \quad (33)$$

$$\text{with } r_0 = 2GE_0.$$

Time scale

$$t_0 = \frac{r_0}{c} = r_0 = 2GE \quad (34)$$

$$(\text{since } c = 1)$$

.10 Scale Hierarchy

Planck length as reference

$$\ell_P = \sqrt{G} = 1 \quad (\text{in nat. units}) \quad (35)$$

Scale ratio

$$\xi_{\text{ratio}} = \frac{\ell_P}{r_0} = \frac{\sqrt{G}}{2GE} = \frac{1}{2\sqrt{G} \cdot E} \quad (36)$$

For $E \sim 1 \text{ GeV}$:

$$\frac{r_0}{\ell_P} \sim 10^7 \quad (\text{sub-Planck}) \quad (37)$$

Particle	Energy [MeV]
Electron	0.511
Muon	105.658
Tau	1776.86

.11 Particles as Field Excitations

Classification by energy

Antiparticles

Negative field excitations: $E_{\text{field}} < 0$

.12 Lepton Masses

The T0 model predicts lepton masses (derivation in Document 003):

Lepton	T0 [MeV]	Exp [MeV]	Δ [%]
Electron	0.507	0.511	0.87
Muon	103.5	105.7	2.09
Tau	1815	1777	2.16

.13 Anomalous Magnetic Moments

Definition

Magnetic moment:

$$\mu = g \cdot \frac{e}{2m} \cdot \frac{\hbar}{2} \tag{38}$$

Anomalous magnetic moment:

$$a = \frac{g - 2}{2} \tag{39}$$

T0 prediction formula

$$a_{\ell} = \frac{\xi}{2\pi} \left(\frac{E_{\ell}}{E_e} \right)^2 \tag{40}$$

Muon

$$\frac{E_\mu}{E_e} = \frac{105.658}{0.511} = 206.768 \quad (41)$$

$$a_\mu = \frac{1.3333 \times 10^{-4}}{2\pi} \times (206.768)^2 \quad (42)$$

$$= 2.122 \times 10^{-5} \times 42,753 \quad (43)$$

$$= 1.166 \times 10^{-3} \quad (44)$$

Electron

$$a_e = \frac{\xi}{2\pi} = 2.122 \times 10^{-5} \quad (45)$$

Tau

$$a_\tau = \frac{\xi}{2\pi} \left(\frac{1776.86}{0.511} \right)^2 = 1.28 \times 10^{-3} \quad (46)$$

.14 Three Field Geometries

Type 1: Localized spherical

$$E(r) = E_0 \left(1 - \frac{\beta}{r} \right), \quad \beta = r_0 \quad (47)$$

Application: Individual particles (electron, muon, tau)

Type 2: Localized non-spherical

$$E(\vec{r}) = E_0 \left(1 - \frac{\beta_{ij} r_i r_j}{r^3} \right) \quad (48)$$

Application: Bound systems

Type 3: Extended homogeneous

Effective parameter:

$$\xi_{\text{eff}} = \frac{\xi}{2} = \frac{2}{3} \times 10^{-4} \quad (49)$$

Application: Cosmology (see Document 026)

.15 Mathematical Identities

Energy field normalization

$$E_{\text{field}}(\vec{r}, t) = E_0 \cdot f(\vec{r}, t) \cdot e^{i\phi(\vec{r}, t)} \quad (50)$$

with:

- E_0 : Characteristic energy
- $f(\vec{r}, t)$: Normalized profile
- $\phi(\vec{r}, t)$: Phase

Duality consistency

Time-mass (Document 003): $T \cdot m = 1$

Time-energy (this document): $T \cdot E = 1$

In natural units ($c = 1$):

$$E = mc^2 = m \quad \Rightarrow \quad T \cdot m = T \cdot E \quad (51)$$

.16 Dimensional Analysis Verifications

Field equation

$$[\nabla^2 E] = [L^{-2}][E] = [E^2][E] = [E^3] \quad (52)$$

$$[4\pi G \rho E] = [E^{-2}][E^4][E] = [E^3] \quad (53)$$

Characteristic length

$$[r_0] = [2GE] = [E^{-2}][E] = [E^{-1}] = [L] \quad (54)$$

Lagrangian density

$$[\mathcal{L}] = [\xi][(\partial E)^2] = [1][E^2] = [E^2] \quad (\text{correct for Lagrangian density}) \quad (55)$$

Anomalous magnetic moment

$$[a_t] = [\xi] \left[\frac{E^2}{E^2} \right] = [1][1] = [1] \quad (56)$$

.17 Formula Reference

Fundamental equations

$$\text{Duality: } T_{\text{field}} \cdot E_{\text{field}} = 1 \quad (57)$$

$$\text{Wave equation: } \square E_{\text{field}} = 0 \quad (58)$$

$$\text{With sources: } \nabla^2 E = 4\pi G \rho E \quad (59)$$

$$\text{Lagrangian density: } \mathcal{L} = \xi(\partial E)^2 \quad (60)$$

Derived constants

$$\text{Characteristic energy: } E_0 = \sqrt{m_e \cdot m_\mu} = 7.348 \text{ MeV} \quad (61)$$

$$\text{Fine structure constant: } \alpha = \xi(E_0/1 \text{ MeV})^2 \approx 1/137 \quad (62)$$

$$\text{Gravitational constant: } G = \frac{\xi^2}{4m_e} \times \text{factors} \quad (63)$$

Characteristic scales

$$\text{T0 length: } r_0 = 2GE \quad (64)$$

$$\text{T0 time: } t_0 = 2GE \quad (65)$$

$$\text{Planck length: } \ell_P = \sqrt{G} = 1 \quad (66)$$

$$\text{Scale ratio: } \xi_{\text{ratio}} = \frac{1}{2\sqrt{GE}} \quad (67)$$

Prediction formulas

$$\text{g-2 formula: } a_\ell = \frac{\xi}{2\pi} \left(\frac{E_\ell}{E_e} \right)^2 \quad (68)$$

$$\text{Parameter: } \xi = \frac{4}{3} \times 10^{-4} \quad (69)$$

$$\text{Effective parameter: } \xi_{\text{eff}} = \frac{\xi}{2} \quad (70)$$

.18 Numerical Values

Fundamental constants (in natural units)

$$\hbar = 1 \quad (71)$$

$$c = 1 \quad (72)$$

$$\alpha = \frac{1}{137.036} \approx 7.297 \times 10^{-3} \quad (73)$$

$$G = 1 \text{ (numerically, dimension } [E^{-2}]) \quad (74)$$

T0 parameters

$$\xi = \frac{4}{3} \times 10^{-4} = 1.3333 \times 10^{-4} \quad (75)$$

$$\xi^2 = 1.7778 \times 10^{-8} \quad (76)$$

$$\frac{\xi}{2\pi} = 2.1221 \times 10^{-5} \quad (77)$$

$$\xi_{\text{eff}} = 6.6667 \times 10^{-5} \quad (78)$$

$$E_0 = 7.348 \text{ MeV (from exp. masses)} \quad (79)$$

$$E_0^{\text{T0}} = 7.398 \text{ MeV (theoretical)} \quad (80)$$

Lepton energies

$$E_e = 0.511 \text{ MeV} \quad (81)$$

$$E_\mu = 105.658 \text{ MeV} \quad (82)$$

$$E_\tau = 1776.86 \text{ MeV} \quad (83)$$

Energy ratios

$$\frac{E_\mu}{E_e} = 206.768 \quad (84)$$

$$\frac{E_\tau}{E_e} = 3477.2 \quad (85)$$

$$\frac{E_\tau}{E_\mu} = 16.817 \quad (86)$$

.19 Calculation Examples

Muon g-2

Given:

- $\xi = 1.3333 \times 10^{-4}$
- $E_\mu = 105.658 \text{ MeV}$
- $E_e = 0.511 \text{ MeV}$

Calculation:

$$\frac{E_\mu}{E_e} = \frac{105.658}{0.511} = 206.768 \quad (87)$$

$$\left(\frac{E_\mu}{E_e}\right)^2 = 42,753.3 \quad (88)$$

$$\frac{\xi}{2\pi} = \frac{1.3333 \times 10^{-4}}{6.2832} = 2.1221 \times 10^{-5} \quad (89)$$

$$a_\mu = 2.1221 \times 10^{-5} \times 42,753.3 \quad (90)$$

$$= 1.1659 \times 10^{-3} \quad (91)$$

Fine structure constant

Given:

- $\xi = 1.3333 \times 10^{-4}$
- $E_0 = 7.398 \text{ MeV}$

Calculation:

$$\left(\frac{E_0}{1 \text{ MeV}}\right)^2 = (7.398)^2 = 54.73 \quad (92)$$

$$\alpha = 1.3333 \times 10^{-4} \times 54.73 \quad (93)$$

$$= 7.297 \times 10^{-3} \quad (94)$$

$$= \frac{1}{137.04} \quad (95)$$

$$\text{Experimental: } \alpha_{\text{exp}} = \frac{1}{137.036}$$

Deviation: 0.03%

Characteristic length (electron)

Given:

- $E_e = 0.511 \text{ MeV} = 0.511 \times 1.6 \times 10^{-13} \text{ J} = 8.2 \times 10^{-14} \text{ J}$
- $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
- $c = 3 \times 10^8 \text{ m/s}$

Conversion to natural units:

$$r_0 = 2GE \approx 10^{-28} \text{ m} \quad (96)$$

Planck comparison:

$$\frac{r_0}{\ell_P} = \frac{10^{-28}}{1.6 \times 10^{-35}} \approx 10^7 \quad (97)$$

0.1 Symbol Reference

Symbol	Meaning	Dimension
ξ	Fundamental parameter	1
E_0	Characteristic energy	E
E_{field}	Universal energy field	E
T_{field}	Intrinsic time field	E^{-1}
r_0	T0 characteristic length	$L = E^{-1}$
t_0	T0 characteristic time	$T = E^{-1}$
ℓ_P	Planck length	$L = E^{-1}$
G	Gravitational constant	E^{-2}
α	Fine structure constant	1
a_ℓ	Anomalous magnetic moment	1
E_e, E_μ, E_τ	Lepton energies	E
m_e, m_μ, m_τ	Lepton masses (= E in nat. units)	E
$\mathcal{L}$	Lagrangian density	E^4
$\square$	d'Alembert operator	E^2
ξ_{eff}	Effective parameter ($\xi/2$)	1

0.2 Unit Conversions

Natural $\rightarrow$ SI

$$1 \text{ (Energy)} = 1 \text{ GeV} = 1.6 \times 10^{-10} \text{ J} \quad (98)$$

$$1 \text{ (Length)} = \frac{\hbar c}{1 \text{ GeV}} = 0.197 \text{ fm} \quad (99)$$

$$1 \text{ (Time)} = \frac{\hbar}{1 \text{ GeV}} = 6.58 \times 10^{-25} \text{ s} \quad (100)$$

Standard natural units

In standard convention ($\hbar = c = 1$):

- $\alpha = \frac{e^2}{4\pi\epsilon_0} \approx \frac{1}{137}$ (dimensionless)
- All quantities in powers of energy
- Physical predictions identical to other conventions

0.3 Relations to Other Documents

- **Document 003:** Time-mass duality, foundations, origin of ξ
- **Document 011:** Fine structure constant (geometric derivation)
- **Document 012:** Gravitational constant (systematic derivation)
- **Document 018:** Geometric g-2 formulation (fractal geometry)
- **Document 019:** Lagrangian formulation (quantum field theory)
- **Document 026:** Cosmology ($\xi_{\text{eff}} = \xi/2$)
All formulations are based on $\xi = \frac{4}{3} \times 10^{-4}$.